Modeling Users' Collaborative Behavior With a Serious Game

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***Abstract*—User modeling is an important area in the field of artificial intelligence. Traditionally, user modeling approaches have focused on individual features of users, leaving aside aspects of users as part of a group. However, in enterprise, academic, or political contexts, it is common to find groups of people working together to solve a collaborative task using online collaborative platforms. In this context, counting with a knowledge based on describing the users' behavior when working in a group (i.e., the collaborative profile) is vital for a system aiming at assisting them to enhance their collaborative abilities that might impact the group dynamics. Traditionally, collaborative abilities have been measured by means of questionnaires or by an external human observer. In this work, we propose an approach to build collaborative profiles by the means of observation of users' behavior in a collaborative serious game. The main advantage of our approach is that it avoids external human observers of the group dynamics and the use of self-perception questionnaires. We obtained promising results when comparing the user profiles obtained with our approach with respect to the classical measurement instruments.**

***Index Terms*—Collaborative behavior, serious games, user modeling.**

I. Introduction

A USER profile is a representation or description of different aspects of a user in a computer application. User profiles are used by operating systems, recommender systems, personalized websites, and social networking sites, among other systems. These systems use the user profiles for different purposes including customization, adaptation, and personalization of the system or the services offered by the system. Particularly, we are interested in modeling user profiles describing the users' behavior while interacting with other users to solve a collaborative assignment. This is particularly important in the software creation market, where the increasing complexity of the applications makes it necessary for the coordinated labor of one or several teams to develop any project. Furthermore, the increasing availability of online collaborative tools makes it possible for students living in different places to complete collaborative assignments in online learning platforms. In this context, collaboration is vital [1]–[5] and online platforms allow intelligent systems to automatically learn the collaborative profiles of users by means of the observation of the users' behavior [6]–[12]. Counting with information about a user's collaborative profile enables us to understand phenomena such as leadership, groups dynamics, and the differences in the efficiency of a user in a different team. As a result, we can obtain a description of key aspects about users that may impact the group performance [13], [14].

The traditional method to obtain information about the collaborative profile of a group of users is by means of self-perception questionnaires. With this approach, users must complete the questionnaires indicating different aspects of their teamwork preferences. This method has two main disadvantages: 1) questionnaires are usually too long and hard to fill; and 2) not all users might be able to understand what some questions refer to without assistance. These disadvantages might generate bias in the questionnaire results and might make it difficult to interpret the user collaborative profile.

To overcome the aforementioned disadvantages, in this work, we propose an approach to build a collaborative profile from the observation of the users' behavior in an online collaborative game that tests the users ability to work in teams. The development of serious games to model users has aroused interest in the scientific community, where a number of advantages are recognized over other instruments [15]–[19]. For example, participants are more attracted and engaged because they are playing a game. Furthermore, the use of a game to measure collaborative skills is simpler and at the same time more effective, since we are using a concrete activity rather than questions to evaluate the users' behavior.

Serious games have been widely used both for the analysis and for the training of many kinds of skills (see Section II). Among the main advantages of using games with respect to other approaches, we can mention that games usually have a specific goal that players must achieve to win the game, with a set of predefined and clear rules to play. In video games, users are more motivated and encouraged to act as they really feel to, since they do not suffer from the risks that the real scenario represented by the game can have [17].

The rest of this paper is organized as follows. In Section II, we present some previous approaches to the use of serious games for improving collaborative work. Then, in Section III-A, we describe the game that was implemented as the shared environment to observe the users' behavior when playing in a team. In Section III-B, we describe the theory, in which we base our approach for categorizing the user's collaborative behavior. In

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Section III-C, we describe the expert system designed for building the users' collaborative profile from the observation of their actions in the serious game. Next, in Section IV, we present the experiments performed to validate our approach, describing the data set collected, the methodology used, and the results obtained. Finally, in Section V, we present our conclusions and future lines of research.

II. Related Work

In this section, we describe some previous work related to the use of serious games for improving collaborative work. Linehan *et al.* [20] developed serious games focused on the collaborative decision-making process. The design of the proposed game creates an appropriate environment to promote the participation in the collaborative decision-making process. During the game, a tutor is allowed to monitor the actions flow, mainly in stages where there are large numbers of votes, to identify negative behaviors in individuals and groups, and to intervene by indicating corrections in the players behavior. The authors reported an improvement in the process of making collaborative and integrative decisions during the successive rounds of the game. Besides the game itself, these authors developed a methodology to quantify the collaborative behavior. This methodology evaluates the structure and quality of the group decision-making processes within the game, based on objective observations and analysis of the social network. The methodology was generalized for use in other collaborative environments and the DREAD-ED [21] was developed, which proposes an innovative teaching methodology providing a serious online game that allows its users to train social skills in a virtual environment.

Other works [22], [23] have explored the use of online multiplayer video games to develop leadership skills in a teamwork. Sánchez and Olivares [24] studied the effect of applying a series of learning activities based on mobile serious games (MSG) for the development of problem solving and collaborative skills. For this study, three serious games (MSGs Evolution, BuinZoo, and Museum) were used together by student teams in order to solve problems in a collaborative way.

Feldman *et al.* [25] materialize a set of serious games that allow the automatic construction of the perceptive profile of the users. In this way, the teacher can group together students with a similar type of perception in search of a better teamwork performance.

Ducheneaut and Moore [26], Ducheneaut *et al.* [27], and Szell and Thurner [28] measure the social dynamics of the network constituted by the different relationships between players, focused on the evolution of the network properties. On the other hand, Schrader and McCreery [29] construct profiles in order to quantify players' experiences as they develop the skills required in the course of the game.

On the other hand, SYMLOG has been extensively used for the analysis of group interactions in different scenarios. For example, Bale's model has been used in the study of work meetings, analyzing in detail the different kinds of interactions and generating individual reports about the behavior of participants, including qualitative information about group dynamics that might be used to improve the efficiency of group work [30]. Another work in this context is the Mood Meter System [31], in which a compact graphical notation is developed to represent the fluctuations of the mood of the participants of a group work activity; the quality of the results obtained from the use of this system presents a great dependence on the mapping model that is used to translate participation and moods into SYMLOG' s variables. Losada and Markovitch [32] propose and develop the tool named "GroupAnalyzer," which is a computerized system to code and analyze group meetings. This system consists of two modules: a codification module that allows for the creation of entries in the meeting protocol and provides tools for the manipulation of this protocol, and an analysis module to study these meeting using SYMLOG; this allows for the detection of dysfunctional practices in meetings. Chen [33] used SYMLOG to study cooperative virtual interactions in the Internet, generating diagrams for the analysis of cooperation dynamics in the observed virtual communities. As opposed to the abovementioned approaches using SYMLOG and to the best of our knowledge, there is no previous work in the literature that automatically builds SYMLOG profiles from the interaction of players in an online collaborative game. In addition, the SYMLOG profiles obtained with our approach are built automatically without any human expert intervention.

III. Materials and Methods

In this section, we describe the serious game we implemented, the information about the game that will be recorded and analyzed, and the technique used to build the user's collaborative profiles.

A. Serious Game Environment

In this paper, we propose the use of a collaborative game as the mean for building user profiles based on the observation of the players' actions. We base the development of the game used in our approach on two previous studies: first, the work of Kelly *et al.* [34], which proposes how to create a serious game for teaching, focusing on traditional gaming issues and the integration of learning tools; and second, the work of Zagal *et al.* [35], which examines a collaborative board game, exposing the lessons and difficulties identified for the design of collaborative games. Among the lessons learned, Zagal *et al.* considered that a collaborative game should highlight the competitiveness problems, i.e., introducing a tension between perceived individual utility and utility of the team, allow individual players to make decisions and take actions without the consent of the team, show players the consequences of their decisions, and encourage team members to make altruistic decisions, that is, to grant different capacities or responsibilities to players. Regarding the difficulties, according to Zagal *et al.*, a collaborative game should: 1) prevent the game from degenerating into one of the players who make the decisions for the team; 2) be attractive, the players must worry about the result and how to make it satisfactory; and 3) be pleasant, the experience has to be different each time and the challenge presented has to evolve.

In the implemented game, the players integrate a team of two to five players, each with a different color chip. The mission is to bring a precious object to home, which is the last stage

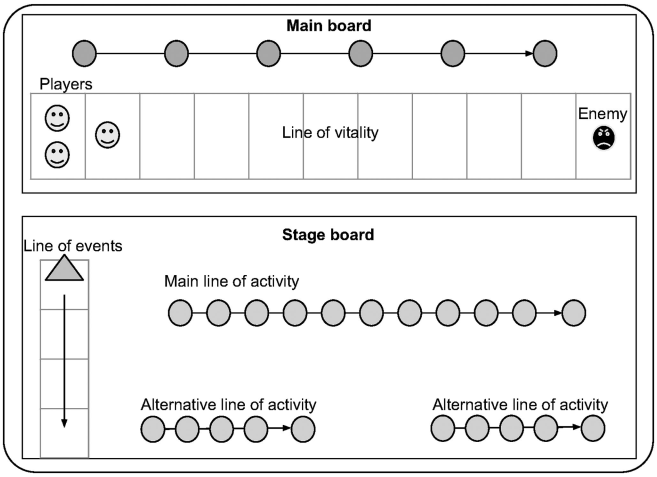

Fig. 1. Scheme of the different boards of the collaborative game.

of the game, avoiding it getting into the enemy hands. Along the way, the team faces several obstacles. The decisions that the players have to take throughout the game aim at overcoming these obstacles to reach the last stage and fulfill the mission. Regardless of whether the team is successful or not, they all receive the same score at the end of the game, which reflects how much progress has been achieved in the trip. The game has two basic components (see Fig. 1). The first component is the main board, which accounts for the general game and shows the level of vitality of each player with respect to the enemy (represented by a black chip) in the line of vitality. The line of vitality is a path composed of several cells, and the player chips and the enemy chip start on opposite sides of this path. Loosing vitality throughout the game causes players to approach the opponent, and vice versa. The second component is a stage board, which contains the events and elements corresponding to a specific part of the path. The stage board is composed of a line of events, composed of cells containing different game actions that players must carry out to fall into that cell and different lines of activity. In each of these lines, there is a marker, which represents the progress of all players along that particular stage of the game. If the enemy reaches any player on the line of vitality at any point in the game, that player is eliminated from the session (he/she "dies"). During each round of the game, the precious object is rotated between players, who are designated, in rounds, responsible of the precious object. If the enemy reaches a player who is holding the precious object, the game is lost for all players.

Each user is given an alias that maintains the anonymity of the participants to avoid that the performance of the participants is influenced or conditioned by the identities of the rest of the team members.

At the beginning of the game, each player obtains a set of cards. Each card represents an action to be done or it contains a symbol associated with one of the lines of activity and allows the player to move along that line. Moreover, at the beginning of each round, each player must draw an activity card from a general stack. These cards can have positive results (such as advancing in a line of activity, allowing players to progress and gaining resources) or negative results (requiring to sacrifice certain resources, moving toward the opponent, or triggering event in the associated line of events).

Both cards and events can cause players to advance toward the enemy or vice versa. After drawing an activity card from the general stack, the player can choose to step back in the line of vitality, to draw two cards from the general stack, or to play up to two cards with symbols associated with any of the lines of activity of the current stage (which allows them to advance in the line). Progress in these lines allows players to get closer to their ultimate goal, while providing them with certain vital resources that would help them in the future. When players reach the end of the main line of activity (there is one per scenario), or the end of the line of events (whichever comes first), they move on to the next stage.

The game also incorporates a voting mechanism for situations in which the game action involves more than one player (for example, one player must be selected by the complete team to play one of his/her cards). This voting mechanism allows us to proceed with the game according to the course of action, which has more consensus among the participants. The game also offers the players a chat so that they can discuss the game situation after coming up with a decision. Players must constantly communicate to share their decisions and the resources they have. The lack of collaboration and agreement between players invariably results in defeat. For example, players cannot display the cards they hold in their hand, but they can talk freely about them using the chat window. Finally, the game has resources to test both collaborative and "selfish" users, that is, players that tend to benefit themselves at the expense of the others.

The game was designed and implemented so that it could be accessible with any web browser, with no need to install anything on the user's computer.[1] During a game, all the players' actions that generate a change in the state of the game are recorded, along with the will of the player/s performing the action.

The developed game facilitates collaboration by promoting both cognitive and organizational skills. The turn-based activities enable users to participate independently to identify their own needs in the game, as well as to manage the available resources. In terms of collaboration, the game promotes users to be active and reflexive, while, at the same time, they are playing to achieve victory of the group as a whole. By using the chat provided by the game, users practice communicative skills, such as asking for information, giving opinions, expressing support, and encouraging others. These items correspond to the classification of collaborative skills proposed by Soller [36].

B. Categorization of the Groups Dynamics

As we previously mentioned, our approach models users by means of the observation of the actions they perform in a collaborative serious game. In the study of groups dynamics, an interaction occurs when an individual performs an action that acts as a stimulus of a response generated by another individual, and vice versa. Each action of our game potentially constitutes an observable interaction. All the actions of the members of a

[1] The game can be accessed at https://collaborative-serious-game.herokuapp.com/

team can be analyzed in relation to their impact on the other members. This interaction process of a group in the search for a common goal is known as group dynamics. The recording of relevant data on the collaboration of each user, based on the observation and analysis of their interactions, allows inferring valuable information about a user's profile. In addition, it is possible to study how to increase the quality of the group dynamics by taking measures that lead to the improvement of those specific characteristics for which the user presents an inadequate degree.

Interaction process analysis (IPA) [13] is among the first research aiming at studying group interactions. Many other coding schemes for sequence interactions were developed since then, for example, SYMLOG [3], [14], CCS [37], [38], or act4team [39], [40]. The multiple-level group observation system (SYMLOG) [3], [14], [41] was developed as an alternative to the limitations of IPA in the sense that IPA does not sufficiently consider nonverbal behavior. SYMLOG simultaneously encodes communication events and content. Basically, by using a questionnaire, a SYMLOG evaluation results in a series of numerical values that consider three dimensions, each with a positive side and a negative side, indicating the presence of different traits that are relevant for group performance. Each dimension is defined as follows.

1) *Upward/downward:* An individual characterized as upward is active and dominant in his/her actions. An individual characterized as downward is relatively quiet and submissive to the dominant members.
2) *Positive/negative:* An individual characterized as positive agrees with other members and is kind when listening. An individual characterized as negative is critical and does not listen to others.
3) *Forward/backward:* An individual classified as forward is self-controlled and has his/her attention placed on the group task. An individual characterized as backward expresses emotions and is not directly interested in the group's main task.

Given the SYMLOG theory, we need to design the way in which the game actions will be translated to the model. In its original form, SYMLOG measurement instrument is a questionnaire consisting of 26 questions that evaluate the different combinations of positive and negative values for each of the three dimensions. The result of this questionnaire is a triple of numerical values, each of them in the range $[-18, 18]$. These three values locate the subject in the 3-D space of the model that characterizes his/her collaborative profile.

A disadvantage of questionnaires as a measurement instrument is that they reflect the user profile in a given moment. However, users' behavior might be different according to the group in which he/she is involved, and moreover, users learn to collaborate over time as they are involved in collaborative activities. By using a collaborative game, we overcome this problem, since the user will participate in different rounds, with different partners, and the profile will evolve reflecting his/her behavior in the long term.

Our approach consists of an expert system that maps the actions performed in the collaborative game to different aspects of the SYMLOG model.

Not all the actions performed by players necessarily provide valuable information for the construction of the profiles, i.e., they do not necessarily provide information about the players. For example, there is an action that moves the enemy by chance; although the previous actions that may have led to this movement may be significant, this action itself does not provide any relevant information about the players.

On the other hand, many actions require players to make a decision in a given moment, and other actions may arise spontaneously from players and provide valuable information about them (for example, a player chooses to play a particular special card at any point in the game). In addition, there are actions that require a general consensus among all players, with the voting mechanism described before.

Another relevant issue considered in the game is the context in which a player chooses between several possible actions. For example, the situation where a player decides to go back on the line of vitality if he/she is only one cell away from the enemy is different to the situation in which the player is far away from him. The state of game at the moment a player makes a decision should be considered because it gives very important information about what kind of behavior a player is demonstrating when making a decision. This requires that the expert system considers the game state when analyzing a game action. By taking all these issues into account, it is possible to associate with each relevant game action and according to the type of action, the context in which it is performed a trinomial of SYMLOG values.

At this point, it is important to note that the result of the SYMLOG questionnaire is calculated on a fixed interval (from -18 to 18), and that all combinations of positive and negative values for each dimension are considered equally in the questionnaire. This fact does not occur in our expert system since not all combinations of values are evaluated uniformly. In the questionnaire, the questions are asked in order to be able to evaluate all possibilities uniformly. In the game, there are a series of predefined actions to which SYMLOG values are associated. Invariably, some dimensions are considered more than others, and the condition of "fixed interval" mentioned above is lost. In addition, only a few possibilities are considered according to the context in which each decision was taken. Therefore, to present the results corresponding to a user, it is necessary to make a normalization according to the ranges in which each dimension has been analyzed.

Finally, it is worth noting that in the methodology proposed by the SYMLOG questionnaire, each user will obtain only a triple as a result of the questions answered. Then, the questionnaire only captures the characteristics of the person at the time it is completed. However, in the game, this triple is generated for each user and each game session. Since it is possible for a user to participate in more than one game session, all the available information is incorporated into his personal profile. Thus, the information contained in each player's profile is variable and reflects the evolution of the player's collaborative profile. As a result, a game session in which a player has performed 100 relevant interactions has much more weight in the analysis than a game session, in which only five interactions are recorded. In order to build a user's profile from many game sessions, the aggregation of many triples (each corresponding to a different game session) into a single one (that identifies the user's collaborative profile) must take these differences into account. Following the example, and having only those two games rounds, the aggregated triple must be more similar to that of the game where the user performed 100 interactions, since it has 20 times more information than the other.

C. Learning User Profiles From the Game Actions

To build user profiles from the use of a serious collaborative game, each action must be associated with a collaborative attribute. The characterization is computed from all the relevant interactions of each user in all the game sessions. In a classical scenario, the association of each action in a game with a collaborative attribute would be performed by a human expert. This task is extremely difficult and time consuming. For this reason, an expert system that automates this procedure is desirable.

An expert system is computer program that uses a knowledge base to infer and solve complex problems. The term is used to refer to systems where knowledge is explicitly represented by related instruments, such as ontologies and rules, rather than incorporating it into the code. An expert system has two types of subsystems: a knowledge base and an inference engine. The knowledge base represents facts about the world. The inference engine represents logical assertions and conditions about the world, generally represented by IF–THEN rules. As a branch of artificial intelligence, expert systems are a type of software system that emulates the decision-making ability of a human expert within a particular domain.

Then, all actions taken by users in a game session will be registered in a game log. The information that is observed from the game sessions includes the actions that each user executes in the regular course of the game, the spontaneous decisions, the participation in collective decision-making situations, and the state of the game variables that condition the performed action. The expert system interprets these game actions as demonstrations of SYMLOG attributes, considering the variant nature of the actions and the context of the game. That is, the game actions differ from each other in form and decision alternatives for the players. This categorization defines three basic types of action.

1) Simple decisions correspond to actions that put a player in the situation in which he/she has to make a decision that can affect to the whole team. In this case, the action selected from the set of available options is evaluated.
2) Spontaneous decisions represent decisions that are not posed to the player as mandatory, but the player decides to perform some action. In this case, we consider that this conduct is worthy of being included in the analysis.
3) Voting decisions correspond to decisions that are taken collaboratively with the voting system of the game. In decisions of this type, we mainly considered the interaction between the users.
   a) Which players are involved in the actions under discussion?
   b) Whether the actions under discussion are positive or negative?
   c) Whether or not an agreement was reached?
   d) What was the vote of each player?
   e) The context of the game in which the situation arose.

During the processing of the game sessions, the inference engine determines which collaborative features are present for each player and provides an assessment. The rules used for mapping user actions to SYMLOG attributes follow the following structure:

$$< \text{user} > \wedge < \text{action} > \wedge < \text{state} > \Rightarrow < \text{SYMLOG} > .$$

The term $<$ user $>$ refers the user who performed the action defined in the term $<$ action $>$ in a given state of the game, represented by the term $<$ state $>$. The available actions in different states of the game include using a card, rolling the dice, taking a new card, discarding a card, moving forward in the line of activity, etc. The state of the game includes information regarding the current stage of the game in the main board and the source that triggered the event (line of events or an activity line). The term $<$ SYMLOG $>$ includes three values representing the way in which the user's decision in the given state describes his/her collaborative profile in each SYMLOG dimension.

The first mapping model was improved after initial configuration experiments, which implied a deep analysis of the actions of the game and their correspondence with the 26 items considered by the SYMLOG questionnaire. After the initial experiments, the necessity of considering the context in which users perform different actions arose. Then, in a second mapping model, we included a new term corresponding to the context in which users take decisions. In the context-aware model, each rule has the following structure:

$$\begin{aligned} &< \text{user} > \wedge < \text{action} > \wedge < \text{state} > \wedge < \text{context} > \\ &\Rightarrow < \text{SYMLOG} > . \end{aligned}$$

The context of the game includes information about the possible actions that the user was able to perform in that given moment and information about the status of the game from the point of view of the user, such as distance to the enemy, position of other players on the board, number of cards at hand, etc.

Rules were designed in such a way that SYMLOG values of different rules can be aggregated to compute the final user profile. This enabled us to build incremental profiles by considering different game sessions in which users might behave differently under the hypothesis that the aggregation of all the game sessions will reflect the user's actual collaborative profile. In consequence, each user profile is incrementally updated after each game session, if sufficient information about his/her participation was recorded. That is, we compute the amount of information available relative to the entire game session. Finally, the profile generated based on the user's participation in the game is presented to the user on a final screen of the site.

IV. Experiments

In this section, we detail the experiments performed to evaluate our approach.

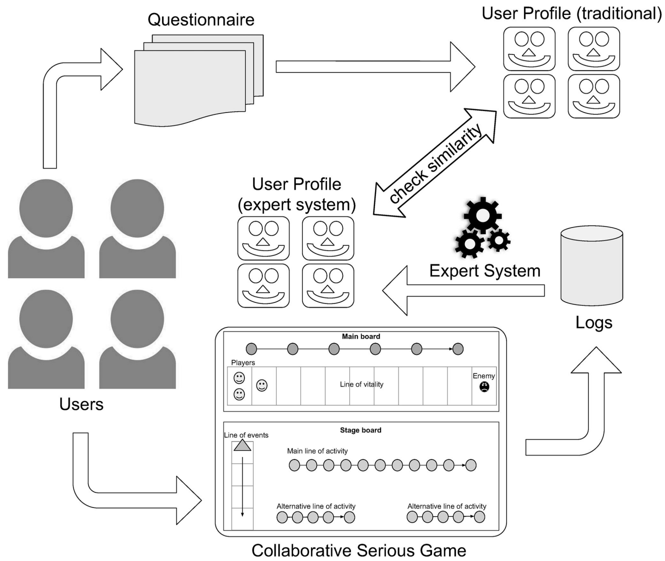

Fig. 2. Proposed approach scheme.

A. Data Set

The utilized data set was collected from students participating in a Systems Engineering course at the National University of the Center of the Province of Buenos Aires, Argentina. Prior to the experiment, students received an introduction to collaborative work and to the SYMLOG questionnaire with the aim of reducing any bias in their "theoretical" profiles due to doubts regarding how to answer the questions. A total of 96 students participated in the experiments, organized in groups of four to five people. Students ages ranged between 20 and 25 years; 81% were males and 19% were females.

Students were free to play as many times game sessions as they wanted. At the end of the experiment, the system registered 385 game sessions. Each game session was stored in a database, registering the identifier of the game session, the identifiers of the users involved in the game session, a flag indicating if the game session was finalized, the result of the game session (victory or defeat), in case of a defeat, the reason, the final score obtained by the team, and the movements or actions performed in the game.

B. Methodology

The methodology of our approach is depicted in Fig. 2. Users were able to access the game platform with any web browser. After a registration process, and only for the first time, students are presented with a SYMLOG questionnaire to obtain his/her self-perception collaborative profile in the traditional way. After answering the questionnaire, users were enabled to play as many game sessions as they desired.

The collected data were processed in two stages: synthesis and discretization. The first stage of the process consisted of synthesizing the collected data to obtain a single record containing the information of the student's interaction in the game, that is the SYMLOG triple. In this stage, for each game session, a normalized partial profile $\mathrm{npp}(U_i, d_j)$ for each user U_i is generated with normalized values for each dimension $d_j =$ *U/D, P/N, F/B* by using

$$\mathrm{npp}(U_i, d_j) = \frac{\sum_{\text{situation}} (v_{i,j} - v_{\min_j})}{\sum_{\text{situation}} (v_{\max_j} - v_{\min_j})}. \quad (1)$$

Each action performed in a given context (i.e., a situation) has an associated triple corresponding to the values in the three SYMLOG dimensions. By a user basis, the values corresponding to all the situations that occurred in each game sessions are aggregated, and the maximum and minimum values, with respect to the possible actions per dimension, are calculated. The term $v_{i,j}$ in (1) corresponds to the value obtained for the dimension d_j, in each situation in which user U_i took a decision, and the terms $v_{\min_j}$ and $v_{\max_j}$ correspond to the minimum and maximum values of the available actions in the given situation, for dimension d_j.

Then, for the construction of the normalized complete profile, the normalized data of a new partial profile $\mathrm{npp}(U_i, d_j)$ are aggregated with the stored profile $\mathrm{ncp}_{\text{prev}}(U_i, d_j)$

$$\mathrm{ncp}(U_i, d_j) = \frac{\mathrm{npc}_{\text{prev}}(U_i, d_j) * |\text{situation}_{\text{prev}}|}{|\text{situation}_{\text{prev}}| + |\text{situation}|} + \frac{\mathrm{npp}(U_i, d_j) * |\text{situation}|}{|\text{situation}_{\text{prev}}| + |\text{situation}|}. \quad (2)$$

To measure the effectiveness of the proposed approach, we will contrast the final user profiles $\mathrm{ncp}(U_i)$ with the profile that resulted from the SYMLOG questionnaires that the users fulfilled after registration in the game's website. The validity of SYMLOG measurement instruments has already been discussed in the literature and this discussion is out of the scope of our paper. We considered the collaborative profiles given by the SYMLOG questionnaires as the ground truth. For this reason, the collaborative profiles obtained with our expert system were contrasted with the profiles obtained by SYMLOG questionnaires, assuming that they represent the actual user collaborative profiles. Additionally, we will compare the impact of considering the context in which the players take decisions, with respect to a direct mapping of the actions to SYMLOG triples without considering the context.

The research questions that we will try to answer with the experiments are the following. 1) Do the profiles generated by the expert system accurately represent the user's collaborative behavior? 2) Is it possible to automate the modeling of the users' collaborative behavior by observing the team dynamics and imitating the SYMLOG measuring instruments? To answer these questions, the profiles of the participants were modeled using the proposed profile construction approach. The resultant profiles were then compared with the profiles obtained with the questionnaire that the participants answered when registering in the game for the first time. In this way, we tried to demonstrate how similar are the profiles built with both approaches.

C. Results

To measure the effectiveness of the profiles obtained with the proposed approach, we take the profiles built with the traditional questionnaire as the ground truth and compute the classical metrics of precision, recall, and accuracy. Tables I–III show the results of these metrics for each dimension of the SYMLOG profiles.

TABLE I
UP/DOWN CONFUSION MATRIX

Real profile →	Down	Neutral	Up	Precision
Down	1	4	0	0.2
Neutral	6	85	0	0.93
Up	0	0	0	-
Recall	0.14	0.96	-	Accuracy: 0.90

TABLE II
POSITIVE/NEGATIVE CONFUSION MATRIX

Real profile →	Negative	Neutral	Positive	Precision
Negative	0	1	0	0
Neutral	0	8	6	0.57
Positive	0	30	51	0.63
Recall	-	0.21	0.89	Accuracy: 0.61

TABLE III
FORWARD/BACKWARD CONFUSION MATRIX

Real profile →	Backward	Neutral	Forward	Precision
Backward	0	3	0	0
Neutral	2	69	22	0.74
Forward	0	0	0	0
Recall	0	0.96	0	Accuracy: 0.72

We can see in Table I that we obtained an accuracy of 90% for the UP/DOWN dimension. However, most of the users are concentrated in the neutral zone of the dimension. To interpret these values, we recall that an individual characterized in the upward dimension is active and dominant in his actions, and individuals characterized as downward are relatively quiet and submissive to the dominant members. In the neutral zone, we find individuals who present an intermediate behavior. For this dimension, most of the students who answered the questionnaire obtained a neutral profile, and by means of the proposed expert system, we identified 96% of these profiles, with a precision of 93%.

For the positive/negative dimension, our approach obtained an accuracy of 61%. None of the students obtained a negative profile according to the questionnaire, and accordingly, no negative profiles were detected by the expert system. A total of 57 students obtained a positive SYMLOG profile, 89% of which were detected by our approach (the remaining 11% was classified as neutral). We recall that an individual characterized as positive tends to agree with other members, while an individual characterized as negative is critical and tends not to listen to the rest.

In the forward/backward dimension, our approach correctly classified 72% of the participants. Most of the participants also had a neutral profile in this dimension, according to the questionnaire. Our approach correctly identified 96% of these users.

In summary, the experiments revealed that 1) the proposed approach to model the user's collaborative profile by observing the user's behavior in a serious game achieves an overall accuracy of 90% in the up/down dimension, 61% in the positive/negative dimension, and 72% in the forward/backward dimension; and 2) user modeling using the proposed approach

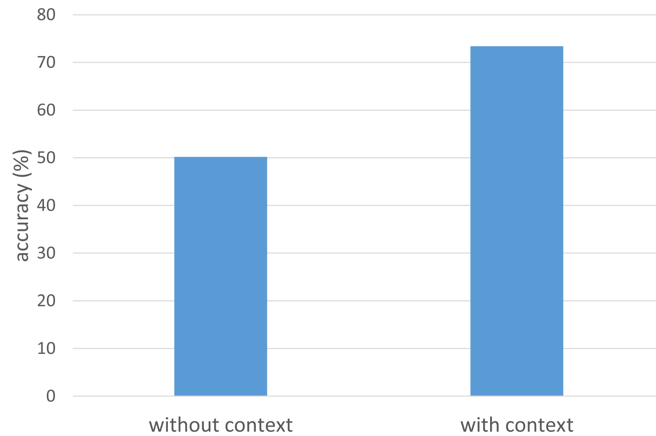

Fig. 3. Influence of the context in the performance of the approach.

resulted in high levels of approximation to the profiles built with the traditional measurement instrument. The accuracy is greater than 60% for the three dimensions, with an average of 73.4%, which we believe to be a reliable value to approximate the user's collaborative profile with respect to the traditional questionnaire.

Finally, we measured the impact of considering the context when classifying the user's actions in the accuracy of the profiles built by the expert system. For this, we grouped all the situations considered in (1) by considering only the action that was performed and not the context. Then, we build the normalized complete user profiles by using (2). Fig. 3 shows the comparison of the accuracy obtained with this simpler approach compared with the proposed approach.

The presented approach achieved 73.4% accuracy, an improvement of 23.2% with respect to the simpler approach that does not consider the context in which users take decisions in the game.

V. CONCLUSION

In this paper, we presented experimental results of an approach to model the collaborative behavior of users by observing their participation in a serious game. The resulting values of accuracy with respect to the traditional measurement instruments (questionnaires) allowed us to determine that the proposed approach is viable to replace the tedious questionnaires with an approach more appealing to users.

We understand that the use of games offers a more enjoyable experience for users in contrast to other platforms and collaboration scenarios. At the same time, the direct observation of the users' behavior offers better fidelity to capture the features needed when building profiles. However, the SYMLOG instruments define extreme profiles that are difficult to find in a collaborative environment as that imposed by the game.

In the literature, there are some studies on the use of serious games in learning environments and for modeling users by considering their individual behavior, but few studies consider the group dynamics and use a collaborative game as the observation environment. Those who do consider the group dynamics with the SYMLOG model require human intervention, which is an expensive and tedious task. For this reason, we believe that our work makes an important contribution to the area of user modeling and the study of group dynamics with serious games.

Future work will focus on designing and analyzing other possible games, incorporating new factors that can positively affect the results, such as the analysis of emotions and roles within the game. Furthermore, we will analyze the use of serious games for training the collaborative skills that users fail to demonstrate according to the profile built with the approach presented in this paper.

References

[1] A. Soller, B. Goodman, F. Linton, and R. Gaimari, “Promoting effective peer interaction in an intelligent collaborative learning system,” in *Proc. Int. Conf. Intell. Tut. Syst.*, 1998, pp. 186–195.

[2] C. Qu and W. Nejdl, “Constructing a web-based asynchronous and synchronous collaboration environment using WebDAV and Lotus Sametime,” in *Proc. 29th Annu. ACM SIGUCCS Conf. User Services*, 2001, pp. 142–149.

[3] R. F. Bales, “SYMLOG: A practical approach to the study of groups,” *Small Groups Soc. Interact.*, vol. 2, pp. 499–523, 1983.

[4] A. Soller, A. Lesgold, F. Linton, and B. Goodman, “What makes peer interaction effective? Modeling effective communication in an intelligent CSCL,” in *Proc. AAAI Fall Symp.: Psychol. Models Commun. Collaborative Syst.*, 1999, pp. 116–123.

[5] S. L. Piezon and R. L. Donaldson, “Online groups and social loafing: Understanding student-group interactions,” *Online J. Distance Learn. Adm.*, vol. 8, no. 4, p. 1, 2005.

[6] A. Gerardo and Y. Yoneo, “Learner models for supporting awareness and collaboration in a CSCL environment,” in *Proc. Int. Conf. Intell. Tut. Syst.*, 1996, pp. 158–167.

[7] Y.-C. Hsu, Y.-H. Ching, and B. L. Grabowski, “Web 2.0 applications and practices for learning through collaboration,” in *Handbook of Research on Educational Communications and Technology*, New York, NY, USA: Springer-Verlag, 2014, pp. 747–758.

[8] K. Apple, M. Reis-Bergan, A. Adams, and G. Saunders, “Online tools to promote student collaboration,” in *Best Practices for Technology-Enhanced Teaching and Learning*, Oxford, U.K.: Oxford Univ. Press, 2011, pp. 239–252.

[9] B. A. Bottge, E. Rueda, J. M. Kwon, T. Grant, and P. LaRoque, “Assessing and tracking students’ problem solving performances in anchored learning environments,” *Educ. Technol. Res. Dev.*, vol. 57, no. 4, pp. 529–552, 2009.

[10] A. Casamayor, A. Amandi, and M. Campo, “Intelligent assistance for teachers in collaborative e-learning environments,” *Comput. Educ.*, vol. 53, no. 4, pp. 1147–1154, 2009.

[11] R. Costaguta, P. Garcia, and A. Amandi, “Using agents for training students collaborative skills,” *IEEE Latin Amer. Trans.*, vol. 9, no. 7, pp. 1118–1124, Dec. 2011.

[12] R. Cheung and D. Vogel, “Activity theory as a design framework for collaborative learning using Google applications technology,” in *Proc. Int. Conf. Web-Based Learn.*, 2012, pp. 140–149.

[13] R. F. Bales, “A set of categories for the analysis of small group interaction,” *Amer. Sociol. Rev.*, vol. 15, no. 2, pp. 257–263, 1950.

[14] R. F. Bales, S. P. Cohen, and S. A. Williamson, *SYMLOG: A System for the Multiple Level Observation of Groups*, New York, NY, USA: Free Press, 1979.

[15] F. L. Greitzer, O. A. Kuchar, and K. Huston, “Cognitive science implications for enhancing training effectiveness in a serious gaming context,” *J. Educ. Resources Comput.*, vol. 7, no. 3, 2007, Art. no. 2.

[16] B. Ives and I. Junglas, “APC forum: Business implications of virtual worlds and serious gaming,” *MIS Quart. Executive*, vol. 7, no. 3, pp. 151–156, 2008.

[17] J. Kirriemuir and A. McFarlane, “Literature review in games and learning,” A NESTA Futurelab Research Rep. 8, 2004.

[18] K. Squire, “Cultural framing of computer/video games,” *Game Stud.*, vol. 2, no. 1, pp. 1–13, 2002.

[19] A. Tang, J. Massey, N. Wong, D. Reilly, and W. K. Edwards, “Verbal coordination in first person shooter games,” in *Proc. ACM Conf. Comput. Supported Cooperative Work*, 2012, pp. 579–582.

[20] C. Linehan, S. Lawson, and M. Doughty, “Tabletop prototyping of serious games for ‘soft skills’ training,” in *Proc. IEEE Games Virtual Worlds Serious Appl.*, 2009, pp. 182–185.

[21] N. Haferkamp, N. C. Kraemer, C. Linehan, and M. Schembri, “Training disaster communication by means of serious games in virtual environments,” *Entertainment Comput.*, vol. 2, no. 2, pp. 81–88, 2011.

[22] M. Popescu, M. Romero, and M. Usart, “Using serious games in adult education serious business for serious people—The MetaVals game case study,” in *Proc. 7th Int. Conf. Virtual Learn.*, 2012, pp. 125–134.

[23] P. Hakkinen, J. Bluemink, M. Juntunen, and I. Laakkonen, “Multiplayer 3D game in supporting team-building activities in a work organization,” in *Proc. IEEE 12th Int. Conf. Adv. Learn. Technol.*, 2012, pp. 430–432.

[24] J. Sánchez and R. Olivares, “Problem solving and collaboration using mobile serious games,” *Comput. Educ.*, vol. 57, no. 3, pp. 1943–1952, 2011.

[25] J. Feldman, A. Monteserin, and A. Amandi, “Detecting students’ perception style by using games,” *Comput. Educ.*, vol. 71, pp. 14–22, 2014.

[26] N. Ducheneaut and R. J. Moore, “The social side of gaming: A study of interaction patterns in a massively multiplayer online game,” in *Proc. ACM Conf. Comput. Supported Cooperative Work*, 2004, pp. 360–369.

[27] N. Ducheneaut, R. J. Moore, and E. Nickell, “Virtual “third places”: A case study of sociability in massively multiplayer games,” *Comput. Supported Cooperative Work*, vol. 16, nos. 1/2, pp. 129–166, 2007.

[28] M. Szell and S. Thurner, “Measuring social dynamics in a massive multiplayer online game,” *Soc. Netw.*, vol. 32, no. 4, pp. 313–329, 2010.

[29] P. Schrader and M. McCreery, “The acquisition of skill and expertise in massively multiplayer online games,” *Educ. Technol. Res. Develop.*, vol. 56, nos. 5/6, pp. 557–574, 2008.

[30] F. Pianesi, M. Zancanaro, E. Not, C. Leonardi, V. Falcon, and B. Lepri, “Multimodal support to group dynamics,” *Pers. Ubiquitous Comput.*, vol. 12, no. 3, pp. 181–195, 2008.

[31] G. L. Rein, D. Bruckmayr, and P. Barthelmess, “Dynamic detection of mood propagation in fan groups,” in *Proc. Program. Workshop Corpora Res. Emotion Affect*, 2008, p. 77.

[32] M. Losada and S. Markovitch, “GroupAnalyzer: A system for dynamic analysis of group interaction,” in *Proc. IEEE 23rd Annu. Hawaii Int. Conf. Syst. Sci.*, 1990, vol. 4, pp. 101–110.

[33] L. L.-J. Chen, “Modeling the internet as cyberorganism: A living systems framework and investigative methodologies for virtual cooperative interaction,” Ph.D. dissertation, Dept. Comput. Sci., Univ. Calgary, Calgary, AB, Canada, 1997.

[34] H. Kelly *et al.*, “How to build serious games,” *Commun. ACM*, vol. 50, no. 7, pp. 44–49, 2007.

[35] J. P. Zagal, J. Rick, and I. Hsi, “Collaborative games: Lessons learned from board games,” *Simul. Gaming*, vol. 37, no. 1, pp. 24–40, 2006.

[36] A. Soller, “Supporting social interaction in an intelligent collaborative learning system,” *Int. J. Artif. Intell. Educ.*, vol. 12, pp. 40–62, 2001.

[37] R. F. Bales and F. L. Strodtbeck, “Phases in group problem-solving,” *J. Abnormal Soc. Psychol.*, vol. 46, no. 4, pp. 485–495, 1951.

[38] R. Fisch, “Eine methode zur analyse von interaktionsprozessen beim problemlösen in gruppen,” *Gruppendynamik*, vol. 25, pp. 149–168, 1994.

[39] S. Kauffeld, “Self-directed work groups and team competence,” *J. Occupat. Org. Psychol.*, vol. 79, no. 1, pp. 1–21, 2006.

[40] S. Kauffeld and N. Lehmann-Willenbrock, “Meetings matter effects of team meetings on team and organizational success,” *Small Group Res.*, vol. 43, no. 2, pp. 130–158, 2012.

[41] R. F. Bales, *Social Interaction Systems: Theory and Measurement*, 2nd ed. Piscataway, NJ, USA: Transaction Publishers, 2012.

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